Assessment tool-1 CO-2

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1.A Genetic Algorithm is applied to maximize the function f(x) = x² using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

A Genetic Algorithm is applied to maximize the function

$$f(x) = x^2$$

The initial population is:

$$P = \{2, 4, 6, 8\}$$

Perform the following:

- Calculate the fitness value of each individual.
- Select the best candidates.
- Perform crossover by exchanging the least significant bits (LSB).
- Explain the role of mutation.

Step 1: Initial Population

Individual (x)

2

Individual (x)

4

6

8

Step 2: Calculate Fitness

Fitness Function:

$$f(x) = x^2$$

Individual 1

$$x = 2$$
$$f(2) = 2^2 = 4$$

Individual 2

$$x = 4$$
$$f(4) = 4^2 = 16$$

Individual 3

$$x = 6$$
$$f(6) = 6^2 = 36$$

Individual 4

$$x = 8$$
$$f(8) = 8^2 = 64$$

Step 3: Fitness Table

Individual (x)	Fitness $f(x) = x^2$
2	4
4	16
6	36
8	64

Step 4: Selection

The individuals with the highest fitness are selected.

Individual	Fitness
8	64
6	36

These are the best candidates (parents).

Step 5: Convert to Binary

Use 4-bit binary representation.

Decimal	Binary
8	1000
6	0110

Step 6: Crossover (Exchange Least Significant Bit)

Parent 1

$$1000$$

Parent 2

$$0110$$

Least Significant Bit (LSB):

- Parent 1 LSB = 0
- Parent 2 LSB = 0

Exchange the LSBs.

Since both LSBs are 0, the offspring remain unchanged.

Offspring 1

$$1000$$

Decimal = 8

Offspring 2

$$0110$$

Decimal = 6

Offspring Table

Offspring	Binary	Decimal
Child 1	1000	8
Child 2	0110	6

Step 7: Final Population

Individual	Fitness
8	64
6	36

The crossover did not change the individuals because both parents had the same least significant bit.

Step 8: Role of Mutation

Mutation is a genetic operator that randomly changes one or more bits in a chromosome.

Example

Suppose Child 2 is

$$0110$$

Mutate the last bit:

$$0110 \rightarrow 0111$$

Binary 0111 = Decimal 7

New fitness:

$$f(7) = 49$$

Thus, mutation introduces a new solution that may be better than the previous one.

Advantages of Mutation

- Increases population diversity.
- Prevents the algorithm from getting stuck in local optimum solutions.
- Helps discover better solutions.
- Improves exploration of the search space.
- Enhances the overall performance of the Genetic Algorithm.

Final Answer (Question 1)

Fitness Values

Individual	Fitness
2	4

Individual	Fitness
4	16
6	36
8	64

Selected Parents

- 8
- 6

Offspring After Crossover

- Child 1 = 8
- Child 2 = 6

Mutation

Mutation randomly changes one or more bits to create new solutions, increase diversity, avoid local optima, and improve the effectiveness of the Genetic Algorithm.

2.A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

A perceptron model is used to classify whether a customer will purchase a product.

Given

Advertisement Exposure

$$x_1 = 2$$

Income Level

$$x_2 = 1$$

Weights

$$w_1 = 0.5$$
$$w_2 = 0.3$$

Bias

$$b = -0.4$$

Step 1: Perceptron Formula

The net input is

$$Net = (w_1 x_1) + (w_2 x_2) + b$$

Step 2: Substitute the Values

$$Net = (0.5 \times 2) + (0.3 \times 1) + (-0.4)$$

Step 3: Perform Multiplication

$$0.5 \times 2 = 1.0$$
$$0.3 \times 1 = 0.3$$

Now substitute

$$Net = 1.0 + 0.3 - 0.4$$

Step 4: Calculate Net Input

$$Net = 0.9$$

Therefore,

$$\boxed{Net = 0.9}$$

Step 5: Apply Step Activation Function

The step activation function is

$$Output = \begin{cases} 1, & \text{if } Net \geq 0 \\ 0, & \text{if } Net < 0 \end{cases}$$

Since

$$0.9 > 0$$

Output = 1

Step 6: Interpretation

Output = 1

Therefore,

The customer will purchase the product.

Final Answer (Question 2)

Given

Parameter	Value
Advertisement Exposure (x_1)	2
Income Level (x_2)	1
Weight (w_1)	0.5
Weight (w_2)	0.3
Bias (b)	-0.4

Net Input Calculation

$$Net = (0.5 \times 2) + (0.3 \times 1) - 0.4$$
$$Net = 1.0 + 0.3 - 0.4$$
$$\boxed{Net = 0.9}$$

Step Activation

Since

$$0.9 \geq 0$$

Output = 1

Decision

The perceptron predicts that the customer will purchase the product.